

Zachary Andrews

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TECHNICAL SKILLS

Compounding & Process Chemistry: formulation & mixing chemistry, acid leaching, solvent extraction, crystallization, Buchner (vacuum) filtration, feed & sample preparation, sample & sieve analysis, calcination

Process Improvement & Reliability: continuous improvement, process optimization, root-cause problem-solving, downtime & efficiency analysis, process standardization, quality assurance

Data & Documentation: process data recording & analysis, data-driven diagnostics, SOP & protocol preparation, technical documentation, MS Excel, Python

Automation & Technology: Arduino-based automation & instrumentation; keen interest in IoT, robotics, and line automation

Safety, Sustainability & Collaboration: WHMIS 2015, SDS/SOP interpretation, hazardous-material handling; waste reduction & closed-loop recovery; cross-functional teamwork

EXPERIENCE

Chemical Technician Intern, Process Research

Process Research ORTECH Inc.

Mississauga, ON

Feb 2026 – May 2026

- Operated, monitored, and troubleshoot pilot process equipment (including a 450 °C furnace with scrubber, gas-handling, and recovery systems), applying root-cause problem-solving to keep runs reliable and on-spec across 20+ pilot trials.
- Reduced waste and improved efficiency by condensing and recycling HCl in a closed loop, and designed and executed a resin ion-exchange test that cut iron contamination by 66% (from 50 ppm to 17 ppm).
- Owned process data recording and analysis for monitoring and diagnostics, and prepared technical documentation to verify output against targets.

Fall Term Academic Representative

University of Waterloo, Chemical Engineering

Waterloo, ON

2025

- Represented the first-year Chemical Engineering cohort, building cross-functional partnerships with faculty and turning feedback into clear recommendations that improved course delivery and resources.

Barista

Starbucks Coffee Company

St. Catharines, ON

May 2024 – Jun 2025

- Kept a high-throughput production line moving with a 20+ person team, cutting bottlenecks under time pressure while holding strict quality and safety standards.

PROJECTS

High-Purity Alumina Extraction from Black Shale

Hatch Ltd., Canadian Energy Metals, Saskatchewan Research Council, Process Research ORTECH

2026

- Contributed hands-on to a multi-organization R&D project producing high-purity alumina from black shale via an integrated hydrometallurgical flowsheet (leaching, solvent extraction, crystallization, purification, and calcination).
- Executed core bench operations end to end: feed and sample preparation, acid leaching, solvent extraction, and Buchner (vacuum) filtration, followed by sample analysis and sieve analysis to control feed particle-size distribution.

Soap Formulation & Process Design

Formulated and evaluated a product using chemical-property analysis and controlled, standardized testing, developing repeatable mixing and batch procedures to keep output consistent, and presented findings at a formal design symposium.

CHE 180

Arduino Temperature-Sensing & Automation

Built a Python and Arduino system for live temperature sensing, data acquisition, and logging, a hands-on entry point into the IoT and automation used to monitor production processes.

Python, Arduino

High-Purity Alumina Extraction from Anorthosite Ore

Purified alumina from anorthosite through acid leaching, filtration, sequential calcium-stripping washes, and furnace thermal treatment, improving final product purity.

EDUCATION

University of Waterloo

Bachelor of Applied Science, Chemical Engineering (Co-op)

Waterloo, ON

Sep 2025 – Apr 2030

- GPA: 3.7 / 4.0
- Areas of Interest: Formulation & Process Chemistry, Manufacturing Operations, Continuous Improvement

CERTIFICATIONS

Standard First Aid incl. CPR-C & AED | WHMIS 2015 | DELF B2 (French)